GUIDO M. FAJARDO GABRIE

Baton Rouge, LA · (225) 472-8311 · guidomanfajardo@gmail.com · gfajar1@lsu.edu
linkedin.com/in/guidofajardo · github.com/gfajardo32

SUMMARY

Computer Science student (Software Engineering concentration) who designs, tests, deploys, and maintains software end to end. Sole developer of a production system a business runs on daily, shipped through tagged releases and a live schema migration. Builds across the full stack in Python, Java, C, and JavaScript, from REST APIs over PostgreSQL to automated Linux and Windows infrastructure, with hands-on machine learning and information retrieval work. Comfortable owning ambiguous problems, debugging to root cause, and communicating technical decisions to non-technical stakeholders. Bilingual in English and Spanish.

EDUCATION

B.S. Computer Science, Software Engineering Concentration | Aug 2022 - Dec 2026
Louisiana State University · Academic Scholarship Recipient (Fall 2022, Spring 2023)

- 107 of 120 credit hours completed with 15 more in progress, spanning advanced data structures and algorithm analysis, operating systems, object-oriented design, software quality and testing, and artificial intelligence
- Member, BengalBots-LSU Robotics Organization

High School Diploma, El Buen Pastor Episcopal Bilingual School | June 2022
Bilingual institution (Spanish/English) · School robotics team, 3 years of inter-school competition

TECHNICAL SKILLS

Languages: Python, C, C++, Java, JavaScript, SQL, C#, PowerShell, x86 Assembly
Data Structures & Algorithms: Advanced data structures and algorithm analysis, complexity analysis, AST traversal, symbol tables, hashing, graph and tree structures
Web & Backend: React, Node.js, Express, Flask, REST API design, JWT and bcrypt authentication, role-based access control, input validation, centralized error handling
Databases: PostgreSQL, relational schema design, normalization (1NF-3NF), joins, aggregations, subqueries, stored procedures, query optimization, schema migrations on live data
Unix/Linux & Systems: Linux (Ubuntu, Arch), gunicorn, systemd, Docker, Cloudflare, VPS provisioning and hardening, VirtualBox and VMware automation, parallel VM workloads
Machine Learning & IR: LLM APIs (Claude, Gemini), text embeddings, cosine-similarity retrieval, retrieval-augmented generation, evaluation design, sliding-window classification, pandas, NumPy
Engineering Practice: Git branching and tagging, semantic versioning, release management with rollback, code review, testing and debugging, root-cause analysis, documentation
AI Productivity Tools: Claude Code CLI, GitHub Copilot, Gemini CLI, agent-assisted refactors, reviewing and correcting generated code, verifying behavior with tests before merge
Languages Spoken: English and Spanish (fully bilingual)

PROJECTS & RESEARCH

Grupo GCHN Invoicing Platform, grupogchn.com (Production) | 2025 - 2026
Python · Flask · Linux VPS · gunicorn · systemd · Cloudflare

- Sole developer on the backend and web application running the daily billing operations of a business in Honduras, owned from requirements through design, development, testing, deployment, and ongoing maintenance
- Encoded regulatory business rules in the backend: four mixed tax categories per line item, government authorization codes, sequential document numbering, and compliant fiscal PDF generation
- Implemented role-based access control scoped per company, admin-provisioned accounts with no public registration, and audit logging; passed a full penetration test with zero critical findings
- Debugged and fixed an off-by-one defect in production invoice counter logic, then shipped feature releases through v1.2.0, including a schema migration run against live data
- Owned deployment on Linux: gunicorn under systemd, origin firewall closed behind a Cloudflare Tunnel, tagged releases deployed to versioned release directories with a symlink for instant rollback

Crash Prediction from Windows Event Logs, CSC 4700 Research | 2026
Python · pandas · PowerShell · VirtualBox · LLM Fine-Tuning

- Proposed and am building a machine learning system to predict a Windows crash five minutes ahead from event-log signals, delivered as a research paper and working demo
- Built an automated harness that runs VMs in parallel, restores a snapshot, injects a fault, detects death by heartbeat, and commits harvested logs to a versioned dataset repository
- Wrote the ingestion pipeline in pandas and caught three defects that would have corrupted training: a UTC versus local timestamp mismatch, stale events baked into the snapshot image, and post-crash events leaking the label
- Framed the problem as sliding-window classification, quantified an 83-to-1 class imbalance and chose metrics accordingly; secured a faculty GPU allocation on LSU's SuperMike-III cluster for model training

Retrieval-Augmented Generation over Threat Intelligence Advisories (in progress) | 2026
Python · Gemini API · Embeddings · NumPy · pypdf

- Building natural language question answering grounded in 10 public CISA ransomware advisories: PDF extraction, boilerplate cleaning, 900-character chunks with 150 overlap, embeddings over 515 chunks, cosine-similarity retrieval
- Evaluated a hosted vector database against plain NumPy arrays and chose NumPy, since a vector store added a dependency without solving a real problem at this corpus size

Personal Finance Dashboard | 2026
Node.js · Express · PostgreSQL · React · JWT

- Designed a normalized PostgreSQL schema (users, categories, transactions) and a full CRUD REST API on Express, with bcrypt password hashing, JWT authentication, input validation, and centralized error handling
- Tested every endpoint with curl before writing any frontend code, then built a React client on top of the API with centralized session-expiry handling

Compiler Construction, CSC 4351 | Spring 2026
Java · C · Object-Oriented Design · Semantic Analysis

- Built a multi-pass compiler in a team; owned the semantic analysis pass in Java, covering declarations, statements, conditionals, and type-checking rules
- Implemented symbol table and scope infrastructure as object-oriented components integrated with teammates' lexer, parser, and code generation passes

Supermarket Database System, CSC 4402 | 2025
Python · SQL · Relational Database Design

- Designed a normalized schema for inventory and transaction management and wrote queries using multi-table joins, aggregations, subqueries, and stored procedures

Mom's Fridge, Recipe Management Web Application, CSC 4330 | 2024 - 2026
JavaScript · Node.js · Agile SDLC

- Delivered a web application with a team across 3 milestone phases and 78 commits, following agile SDLC with documented deliverables and team code reviews

WORK EXPERIENCE

Peer Advisor Researcher, LSU Division of Computer Science & Engineering | Sep 2026 - Present
Baton Rouge, LA

- Selected by the division to advise undergraduate computer science students on degree requirements, course sequencing, prerequisites, and registration logistics
- Resolve routine curriculum questions directly against catalog and degree requirements, and escalate degree audits, overrides, and policy decisions to professional advising staff
- Support division research and documentation work, recording advising interactions to surface recurring curriculum and student-experience issues

IT Support Technician | Summers 2022 - Present
Fajardo & Associates Law Office / Lotificacion Miguel Calvo / Inversiones Multiples Urbe

- Sole technical point of contact across three businesses (legal, real estate, retail pharmacy), supporting hardware, software, networks, and point-of-sale systems
- Managed priorities and follow-through with non-technical owners and staff, turning business problems into technical decisions in both English and Spanish

RELEVANT COURSEWORK

Advanced Data Structures & Algorithm Analysis · Operating Systems · Systems Programming · Object-Oriented Design · Database Systems · Software Systems Development · Software Quality & Testing (in progress) · Artificial Intelligence (in progress) · Programming Languages · Compiler Construction · Computer Organization & Design · Discrete Structures · Engineering Statistics